MAE5005 Advanced Computational Fluid Dynamics, Spring 2026

Homework Set 1: Simple codes and computer representation of real numbers

Due Monday, March 9, 2026 by 19:00, by online submission through *Blackboard*

1. (10 points) Provide a face photo of yourself, and write one paragraph to introduce yourself to your instructor.

2. (30 points) In the lecture, we answered the question: If a coin is tossed N times, what is the probability (P) of having the head exactly M times? If the probability of getting head or tail for each toss is the same, then

$$P = \frac{C_M^N}{2^N} = \frac{N!}{M!(N-M)!\, 2^N}, \tag{1}$$

where C_M^N is the binomial coefficient representing the total number of order combinations of having M heads with N tries. You are now asked to plot P as a function of M for $M = 0, 1, 2, ..., N$ for a given N.

   (a) For $N = 6$, find the results by hand calculations for all possible M;

   (b) Next for $N = 600$, write a Fortran program (by extending the short program in class in some way) to obtain the exact results for P as a function of M. Plot the results as a function of M. Note that you will encounter some small numbers, and you may set $P = 0$ if $P < 10^{-6}$.

   (c) Now ask DeepSeek to do Part (a) and Part (b) for you. Propose the proper question for DeepSeek and ask DeepSeek to provide you a Fortran 90 code. Understand and correct the resulting code. Compare the code from DeepSeek with your own code. Is there anything that you learn from interacting with DeepSeek.

3. (15 points) Using the department Unix computer and the Fortran code inputoutput.f90 we used in the lecture, investigate how an input real number is outputted (*i.e.*, interpreted by a computer in bits and bytes) under the single precision representation.

   (a) Input = 0.31E−45,

   (b) Input = 0.82E−45

   (c) Input = 3.2,

   (d) Input = −1234.5,

   (e) Input = 3.41E+38

   Namely, in each case, (i) report the actual output of your input; (ii) explain the precise binary form used by the computer. Your code is expected to be less than 10 lines.

   Next, ask DeepSeek to do this problem for you. Propose the proper question for DeepSeek and ask DeepSeek to provide you a Fortran 90 code. Try to run the resulting code. Will the DeepSeek code provide you the correct results? What do you learn in this exercise?

Hint: you may consult the Wiki page at
https://en.wikipedia.org/wiki/Single-precision_floating-point_format

4. (15 points) As discussed in the class, computer represents real values by binary representation. We have learned that the single-precision representation of a real number makes use of 32 bits (or 4 bytes), where the first bit indicates the sign of the real number, the next 8 bits represents the exponent, and the remaining 23 bits describes the fraction. The conversion from the binary to the real number is done by the following:

$$\text{real number} = \begin{cases} (-1)^{\text{sign}} \times 2^{-126} \times 0.\text{fraction}, & \text{if the exponent is zero (subnormal value)} \\ (-1)^{\text{sign}} \times 2^{\text{exponent}-127} \times 1.\text{fraction}, & \text{if the exponent is non-zero (normal value)} \end{cases}$$

   (a) According to the above conversion relations, what is the exact magnitude of the smallest non-zero normal value? And what is the corresponding binary representation?

   (b) What is the exact magnitude of the largest subnormal value? And what is the corresponding binary representation?

   (c) What is the theoretical round-off error when computing $(\pi^2/12)$ (known as the first Nielsen–Ramanujan constant)?

   (d) What is the theoretical round-off error when computing $\sqrt{2}$?

   (e) From the above, what can you conclude regarding the number of significant digits (in decimal) that a single-precision representation can produce?

Notes: you may use my demo code inputoutput.f90, as needed.